

Bucket Sort: Analysis (1)

BUCKET-SORT(A, n)

let $B[0..n-1]$ be a new array

for $i = 0$ **to** $n - 1$

 make $B[i]$ an empty list

for $i = 1$ **to** n

 insert $A[i]$ into list $B[\lfloor n \cdot A[i] \rfloor]$

for $i = 0$ **to** $n - 1$

 sort list $B[i]$ with insertion sort

concatenate lists $B[0], B[1], \dots, B[n-1]$ together in order

return the concatenated lists

Bucket Sort: Analysis (2)

- Let n_i be the number of elements in $B[i]$.
- Since Insertion-sort has $O(n^2)$ running time for input size of n , we have

$$T(n) = \Theta(n) + \sum_{i=0}^{n-1} O(n_i^2)$$

- Take expectations on both sides:

$$\begin{aligned} E(T(n)) &= \Theta(n) + \sum_{i=0}^{n-1} E(O(n_i^2)) \\ &= \Theta(n) + \sum_{i=0}^{n-1} O(E(n_i^2)) \end{aligned}$$

Bucket Sort: Analysis (3)

Claim: $E(n_i^2) = 2 - 1/n$, for $i = 0, 1, \dots, n - 1$.

Proof: Define indicator random variables

$$x_{ij} = \begin{cases} 1 & \text{if } A[j] \text{ falls in bucket } i \\ 0 & \text{otherwise} \end{cases} \quad \Rightarrow \quad n_i = \sum_{j=1}^n x_{ij}$$

$$\begin{aligned} E(n_i^2) &= E\left[\left(\sum_{j=1}^n x_{ij}\right)^2\right] = E\left[\left(\sum_{j=1}^n \sum_{k=1}^n x_{ij}x_{ik}\right)\right] \\ &= E\left[\left(\sum_{j=1}^n x_{ij}^2\right) + \sum_{j=1}^n \sum_{\substack{k=1 \\ k \neq j}}^n x_{ij}x_{ik}\right] = \sum_{j=1}^n E(x_{ij}^2) + \sum_{j=1}^n \sum_{\substack{k=1 \\ k \neq j}}^n E(x_{ij}x_{ik}) \end{aligned}$$

Bucket Sort: Analysis (4)

- Note that x_{ij} is 1 with probability $1/n$ and 0 with probability $(1 - 1/n)$. Therefore:

$$E(x_{ij}^2) = 1^2 \cdot \frac{1}{n} + 0^2 \cdot (1 - \frac{1}{n}) = \frac{1}{n}$$

- When $j \neq k$, x_{ij} and x_{ik} are independent random variables.

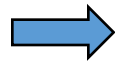
$$E(x_{ij}x_{ik}) = E(x_{ij})E(x_{ik}) = \frac{1}{n} \cdot \frac{1}{n} = \frac{1}{n^2}$$


$$E(n_i^2) = \sum_{j=1}^n \frac{1}{n} + \sum_{j=1}^n \sum_{\substack{k=1 \\ k \neq j}}^n \frac{1}{n^2} = 1 + \frac{(n^2 - n)}{n^2} = 2 - \frac{1}{n}$$

Bucket Sort: Analysis (5)

- Finally we have:

$$\begin{aligned} E(T(n)) &= \Theta(n) + \sum_{i=0}^{n-1} O(E(n_i^2)) \\ &= \Theta(n) + n \times O(2 - 1/n) = \Theta(n) \end{aligned}$$

 the expected running time of bucket sort is $\Theta(n)$.